

D497

Data Wrangling

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

An analyst at a travel startup with 25 staff pulls hotel data from an API. Each hotel record holds a list of rooms, and each room holds a list of nightly rates. The API launched last year. What must she do to analyze rates in a table?

- A. Wait for the API to mature before using it.
- B. Read the response directly as a CSV file.
- C. Keep only the hotel-level fields.
- D. Flatten the nested JSON so each row is one rate with its hotel and room fields.

Target word _____	The one fact that decides _____	Answer ___	☐ sure ☐ unsure ☐ guess
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2.

A customer table has 8,000 rows. The email column has 600 blank values and 200 values that read "N/A." The table was exported from a CRM on a Friday. What percentage of emails are missing?

- A. 2.5%
- B. 7.5%
- C. 90%
- D. 10%

Target word _____	The one fact that decides _____	Answer ___	☐ sure ☐ unsure ☐ guess
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3. [TRANSFER — reversed stem / different variable]

After cleaning, 4% of a 25,000-row column is still missing. How many values are present?

- A. 24,000
- B. 25,000
- C. 24,996
- D. 1,000

Target word _____	The one fact that decides _____	Answer ___	☐ sure ☐ unsure ☐ guess
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4.

A clinic's patient table should have one row per patient_id. A check finds 10,250 rows but only 10,000 distinct patient IDs. The table feeds a dashboard that refreshes at 6 a.m. What should the analyst conclude?

- A. Delete every row whose ID appears more than once.
- B. There are 10,250 patients.
- C. There are 250 duplicate rows; investigate and deduplicate before analysis.
- D. The 6 a.m. refresh created the extra patients.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

In a salary survey from a staffing agency with 11 branches, the highest earners are the most likely to skip the income question. The survey was translated into Spanish. What kind of missing data is this?

- A. Missing at random (MAR).
- B. Missing not at random (MNAR).
- C. Not missing, because blanks mean zero income.
- D. Missing completely at random (MCAR).

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

Which situation describes data missing completely at random (MCAR)?

- A. Any missing value in a large file is MCAR.
- B. Older patients skip an online form more often.
- C. The sickest patients are least likely to report symptoms.
- D. A lab machine randomly fails to record 2% of readings, unrelated to any patient trait or reading value.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A home-price column is strongly right-skewed by a few mansions, and 3% of values are missing. The data comes from a county office that updates monthly. Which simple imputation is most appropriate?

- A. Zero.
- B. The median price.
- C. Wait for next month's county update.
- D. The mean price.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

For which column is mode imputation the appropriate simple choice?

- A. A categorical column such as preferred contact method.
- B. Any column, since the mode is always best.
- C. A heavily skewed income column.
- D. A roughly symmetric height column.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A merged sales file from two regions stores dates as 03/04/2025 in one source and 2025-04-03 in the other. Both regions report to the same vice president. Before analysis, what should the analyst do?

- A. Assume both sources are month-first.
- B. Confirm each source's format and convert all dates to one standard date type, such as ISO 8601.
- C. Sort the dates as text.
- D. Ask the vice president which region is more important.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

Weights arrive in kilograms from one sensor and in pounds from another (1 lb = 0.4536 kg). What is a 110-pound reading on the kilogram scale?

- A.** About 242.5 kg
- B.** 110 kg
- C.** About 49.9 kg
- D.** No change needed; mixed units average out in large samples.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

An analyst at a nonprofit with a \$2 million budget needs daily weather data. The provider offers a documented API limited to 1,000 calls per day, and its terms forbid scraping its web pages. What is the appropriate approach?

- A.** Use the API within its rate limit and terms.
- B.** Scrape the web pages because it is faster.
- C.** Make 5,000 API calls a day to be safe.
- D.** Scrape anyway, since public data can always be collected.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

An analyst cleaned a 40-column survey file by hand-editing cells in a spreadsheet and saved over the original. The file was due to a client on Tuesday. What best practice was broken?

- A.** None, as long as the client received the file Tuesday.
- B.** Manual edits are always fine if the analyst is careful.
- C.** Keep the raw data unchanged and apply cleaning in a documented, rerunnable script.
- D.** Clean in a spreadsheet with macros disabled.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

An analyst at a travel startup ~~with 25 staff~~ pulls hotel data from an API. Each hotel record holds a list of rooms, and each room holds a list of nightly rates. ~~The API launched last year.~~ What must she do to analyze rates in a table?

Target word: wrangling step **Decides:** Rates are nested two levels deep.

A	Camouflage	The API's age does not change the nesting.
B	Confusion set	CSV readers expect flat rows, not nested lists.
C	Dropped element	Throws away the rates she needs.
D	Correct	Normalizing nested lists creates one row per unit of analysis.

2. Answer D Concept 2

A customer table has 8,000 rows. The email column has 600 blank values and 200 values that read "N/A." ~~The table was exported from a CRM on a Friday.~~ What percentage of emails are missing?

Target word: missing rate **Decides:** Both blanks and "N/A" are missing.

A	Wrong stem number	200 / 8,000 counts only the placeholders.
B	Dropped element	600 / 8,000 ignores the "N/A" placeholders.
C	Inverted operation	That is the share present, not missing.
D	Correct	$(600 + 200) / 8,000 = 10\%$.

3. Answer A Concept 2 · transfer

After cleaning, 4% of a 25,000-row column is still missing. How many values are present?

Target word: solve for present count — reverse **Decides:** Present = 96% of 25,000.

A	Correct	$25,000 \times 0.96 = 24,000$.
B	Dropped element	Ignores the missing values.
C	Wrong stem number	Subtracts 4 instead of 4%.
D	Inverted operation	That is the missing count.

4. Answer C Concept 3

A clinic's patient table should have one row per patient_id. A check finds 10,250 rows but only 10,000 distinct patient IDs. ~~The table feeds a dashboard that refreshes at 6 a.m.~~ What should the analyst conclude?

Target word: duplicate count and action **Decides:** $10,250 - 10,000 = 250$ extra rows.

A	Absolute language	Removes the real record along with the copies.
B	Dropped element	Ignores the uniqueness check.
C	Correct	Rows beyond the distinct keys are duplicates to review.
D	Camouflage	Refresh timing does not explain duplicate keys.

5. Answer B Concept 4

In a salary survey from a staffing agency with 11 branches, the highest earners are the most likely to skip the income question. ~~The survey was translated into Spanish.~~ What kind of missing data is this?

Target word: missingness type **Decides:** Missingness depends on the income value itself.

A	Adjacent category	MAR depends on other observed variables, not the missing value.
B	Correct	The chance of missing depends on the unrecorded value.
C	Inverted operation	Treats a skipped answer as a real value.
D	Confusion set	MCAR is unrelated to any value.

6. Answer D Concept 4 • transfer

Which situation describes data missing completely at random (MCAR)?

Target word: example of MCAR — reverse **Decides:** Missingness unrelated to anything.

A	Absolute language	File size does not decide the mechanism.
B	Adjacent category	Related to an observed variable (age): MAR.
C	Inverted operation	Related to the missing value itself: MNAR.
D	Correct	Pure chance drives the gaps.

7. Answer B Concept 5

A home-price column is strongly right-skewed by a few mansions, and 3% of values are missing. ~~The data comes from a county office that updates monthly.~~ Which simple imputation is most appropriate?

Target word: imputation choice **Decides:** Strong right skew.

A	Dropped element	Drops the fact that zero is not a real price.
B	Correct	The median is not pulled by the mansions.
C	Camouflage	The update schedule does not fill the gaps.
D	Confusion set	The mean is inflated by the skew.

8. Answer A Concept 5 • transfer

For which column is mode imputation the appropriate simple choice?

Target word: column type for mode — reverse **Decides:** Mode fits categorical data.

A	Correct	The most common category is a sensible fill.
B	Absolute language	The right choice depends on type and shape.
C	Confusion set	Median fits skewed numbers.
D	Adjacent category	Mean fits symmetric numbers.

9. Answer B Concept 6

A merged sales file from two regions stores dates as 03/04/2025 in one source and 2025-04-03 in the other. ~~Both regions report to the same vice president.~~ Before analysis, what should the analyst do?

Target word: standardization step **Decides:** Two date conventions in one file.

A	Confusion set	Assumes a shared convention that may not exist.
B	Correct	One consistent type prevents month-day mix-ups.
C	Dropped element	Skips conversion, so text sorting scrambles order.
D	Camouflage	Reporting lines do not fix formats.

10. Answer C Concept 6 · transfer

Weights arrive in kilograms from one sensor and in pounds from another (1 lb = 0.4536 kg). What is a 110-pound reading on the kilogram scale?

Target word: unit conversion **Decides:** Multiply pounds by 0.4536.

A	Inverted operation	Divides by 0.4536, converting the wrong way.
B	Dropped element	Leaves the value unconverted.
C	Correct	$110 \times 0.4536 = 49.9$.
D	Absolute language	Mixed units bias every result.

11. Answer A Concept 7

An analyst at a nonprofit ~~with a \$2-million budget~~ needs daily weather data. The provider offers a documented API limited to 1,000 calls per day, and its terms forbid scraping its web pages. What is the appropriate approach?

Target word: gathering method **Decides:** An API exists with stated limits and terms.

A	Correct	The sanctioned channel respects the provider's rules.
B	Adjacent category	A different collection method that the terms forbid.
C	Dropped element	Ignores the 1,000-call limit.
D	Absolute language	Terms of use still apply to public pages.

12. Answer C Concept 8

An analyst cleaned a 40-column survey file by hand-editing cells in a spreadsheet and saved over the original. ~~The file was due to a client on Tuesday.~~ What best practice was broken?

Target word: broken practice **Decides:** Raw data was overwritten and steps were not recorded.

A	Camouflage	Meeting the deadline does not make the work reproducible.
B	Absolute language	Care does not create an audit trail.
C	Correct	Protects the original and makes each step repeatable.
D	Adjacent category	Still manual and unrecorded.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	8	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	5	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	5	
Absolute language	Always, never, guarantee — where judgment applies.	6	
Confusion set	Every option is real; the test is exact role-matching.	5	
Camouflage	Answers a stem fact that was never the question.	5	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Nested vs. flat formats	JSON and XML can nest lists inside records; to analyze them in a table, flatten (normalize) them so each row is one unit of analysis.	Reading nested JSON as if it were a flat CSV.	Records containing lists of other records.
2	Measuring completeness	Count every form of missing value (blanks, NA, placeholder text like “N/A”) before computing a missing rate.	Counting only true blanks.	Placeholder strings mixed with blanks.
3	Duplicates and key uniqueness	If a key should be unique, rows minus distinct keys = duplicate rows; review them before removing.	Counting rows as entities, or deleting every row with a duplicated key.	Row count and distinct-key count differ.
4	Missingness mechanisms	MCAR = missing unrelated to anything; MAR = related to other observed variables; MNAR = related to the missing value itself.	Assuming all missing data is random.	Who is likely to skip the question, and why.
5	Choosing an imputation	Median for skewed numeric data, mean for roughly symmetric numeric data, mode for categories.	Using the mean on skewed data, or zero as a stand-in for unknown.	A skew or data-type description plus a missing rate.
6	Standardizing formats and units	Convert every source to one date format, unit, and type before combining.	Assuming two sources use the same convention.	Two date styles or two units in a merged file.
7	Responsible data gathering	Prefer an official API, stay within its rate limits, and follow the provider's terms.	Scraping pages because the data is public or faster to get.	An available API with stated limits or terms.
8	Reproducible cleaning	Keep raw data unchanged and apply every cleaning step in a documented, rerunnable script.	Hand-editing cells and saving over the original.	Manual edits in a spreadsheet.

Course scope (public description): Data Wrangling elaborates on concepts covered in Introduction to Data Science, helping to develop skills crucial to the field of data science and analysis. It explores how to wrangle data from diverse sources and shape it to enable data-driven applications—a common activity in many data scientists' routine. Topics covered include gathering and extracting data from widely-used data formats, assessing the quality of data, and exploring best practices for data cleaning.